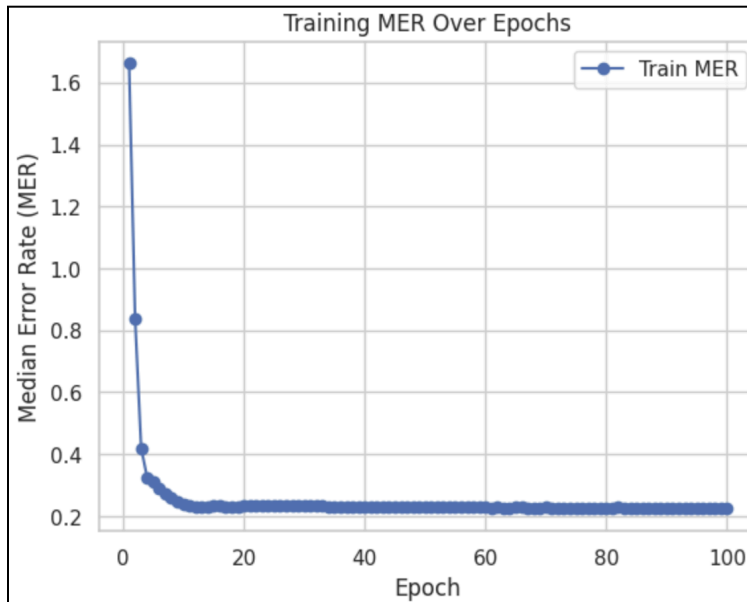


- **Observations and interpretations for plots:**

Plot-1: MER for **linear regression** model



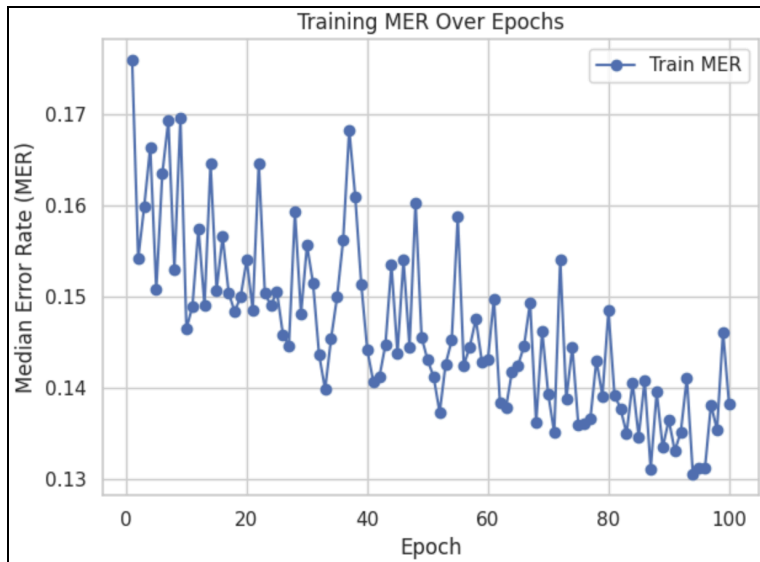
Observation 1: Initial Sharp Decrease in MER

Explanation: This sharp decline suggests that the linear model is quickly capturing the most important linear relationships between the input features and the target variable (sale_price). Also, the initial high MER (~1.6) indicates that the model started with relatively poor predictions, but constant training brought a quick improvement.

Observation 2: Plateau After ~10 Epochs (stabilize at around 0.2)

Explanation: The model has converged to a stable solution, meaning further weight updates are yielding minimal improvement in prediction accuracy. On average, the predicted sale prices are within 20% of the actual prices across the training dataset.

Plot-2: MER for **multi-layer perceptron** model (MLP)



Observation 1: Overall Downward Trend

Explanation: This general downward trend suggests that the model is learning effectively. As training progresses, the MLP is minimizing errors, improving its accuracy on the training data.

Observation 2: Unstable Fluctuations

Explanation: The plot shows significant variability in early epochs as the model undergoes rapid adjustments to its weights. In the mid to late training stage, the periodic increases in error rate could indicate potential problems like overfitting. In addition, there is a flatten trend in these final epochs, showing that the MLP model is no longer learning significantly from the data as before.

- **Question 1:** What are your final training errors of the multilayer perceptron model and the linear regression model?

The final training error rate of the linear regression model is 0.2253, and the final training error rate for the multilayer perceptron model is 0.1382. This suggests that the multilayer perceptron model is more accurate in predicting house sale prices.

- **Question 2:** What is the test error shown on Kaggle? How does it compare with the train error?

The test error shown on Kaggle is 0.14822, which is slightly higher than the train error of 0.1382. This indicates that the MLP model generalizes fairly well to

predict the house sale price. However, the increase in error highlights potential room for model optimization.

- **Insights for senior management at Zillow:**

Based on the two graphs, the MLP outperforms the linear model in reducing MER, highlighting its ability to capture non-linear relationships in Zillow's real estate data.

Zillow's AVM is trained on vast datasets nationwide but our models focus specifically on a localized market (Denver single-family homes and condos) with potentially different features, weights, and patterns that Zillow's generalized AVM may miss. This can offer more accurate, region-specific insights for Denver.

Even the best models like Zestimates have limitations. Zillow might overfit or underfit certain areas or property types. Our models can act as a benchmarking tool to evaluate where their AVM struggles. Zillow could compare the accuracy of our models against Zestimates in Denver to see if there are particular property categories or neighborhoods where Zillow consistently underestimates or overestimates prices. Our model can also provide segmentation insights to optimize buying and selling risks in specific property markets.

There is potential to improve Zillow's iBuying Accuracy. Based on the case, the reason why their iBuying model faced losses is because static valuations don't account for rapidly changing market conditions. However our MLP model can handle non-linear relationships. It can provide real-time updates as market conditions evolve and better capture non-traditional factors (e.g., seasonal fluctuations, macroeconomic data).